

Problem 1: The ages (years) and the logarithmically transformed admission serum urea (natural logarithm $\ln$ urea) for twenty patients attending an accident and emergency unit are given below. Compute the Pearson correlation coefficient to determine whether there is a relationship between the two variables.

Subject	Age	$\ln$ urea	Subject	Age	$\ln$ urea
1	60	1.099	11	72	2.617
2	76	1.723	12	91	2.701
3	81	2.054	13	76	2.054
4	89	2.262	14	39	1.526
5	44	1.686	15	71	2.002
6	58	1.988	16	56	1.526
7	55	1.131	17	77	1.825
8	74	1.917	18	37	1.435
9	45	1.548	19	64	2.460
10	67	1.386	20	84	1.932

Problem 2: In a drawing competition, ten candidates were judged by two judges. The ranks given by them are as follows:

Candidate	A	B	C	D	E	F	G	H	I	J
Rank by Judge 1	1	4	8	6	7	3	2	5	10	9
Rank by Judge 2	2	3	9	6	5	1	4	7	8	10

Compute the Spearman rank correlation coefficient.

Problem 3: The ages (years) and the logarithmically transformed admission serum urea (natural logarithm $\ln$ urea) for twenty patients attending an accident and emergency unit are given below.

Subject	Age	$\ln$ urea	Subject	Age	$\ln$ urea
1	60	1.099	11	72	2.617
2	76	1.723	12	91	2.701
3	81	2.054	13	76	2.054
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The equation of the regression line (calculated using the method of least squares) for the data given above is $\ln \text{ urea} = (0.017 \times \text{age}) + 0.72$. Calculate the predicted $\ln$ urea of a patient aged 60 years.

Problem 4: The interval (in seconds) between male mating calls in the grey tree frog, *Hyla versicolor*, was measured at different temperatures. The equation of the regression line is $\text{interval} = (-0.205 \times \text{temperature}) + 8.36$. Calculate the predicted temperature at which the inter-call interval would be 16 seconds.

Problem 5: The equation of the regression line of the data given below is $y = 1.5429x + 2.0286$. Calculate the coefficient of determination.

x_i	y_i	x_i	y_i
1	2	6	11
3	9	7	14
5	9	9	15

Problem 1: An illustrative data set is given below. Carry out one-way analysis of variance to determine whether the difference between the means of the three groups is significant. (Given: $F_{2,6,0.9} = 3.46$; $F_{2,6,0.95} = 5.14$; $F_{2,6,0.975} = 7.26$; $F_{2,6,0.99} = 10.92$; $F_{2,6,0.995} = 14.54$; $F_{2,6,0.999} = 27.00$)

Group 1	Group 2	Group 3
10	19	14
12	20	16
14	21	18

Problem 1: An experiment is to be conducted to study the flow of suspended particles in a coolant used with industrial equipment. The coolant is to be forced through a slit aperture in the middle of a fixed length of pipe. There are three experimental factors in this study: coolant, a qualitative factor at two levels (1, 2); pipe angle, a quantitative factor at four levels (15, 30, 45, 60 degrees from horizontal); and flow rate, a quantitative factor at two levels (60, 90 ft/sec).

How many test runs will have to be carried out to investigate the effect of the three factors at different levels using the OFAT approach?

Problem 2: A company is evaluating two different word processing packages for use by its clerical staff. Part of the evaluation is how quickly a test document can be entered correctly using the two programmes. Twenty secretaries will take part in the evaluation, and each secretary will enter the document twice, using each programme once.

What are the potential pitfalls in nonrandomized designs of such an evaluation? How can a random design be used to avoid these pitfalls?

Problem 3: How can eight students be randomly grouped into one group of four and two groups of two?

Problem 4: In a laboratory investigation of torque forces experienced by a rotating shaft braced by a stationary cylindrical sleeve lubricated to prevent friction between the shaft and sleeve, the factors and levels involved are: shafts (made of steel or aluminum alloys), sleeves (made of two types of metal, one porous and the other nonporous), and lubricants (from four different manufacturers).

How many factor-level combinations are possible in this investigation? How would one carry out complete randomization of the run order of this experiment?

Problem 5: If two tests for each factor-level combination are to be carried out for the experiment given in Problem 4, how would one carry out complete randomization of the run order?

Problem 6: Construct a table showing the effects representation of three two-level factors in an experimental study.

Problem 7: The factors of interest in a chemical process are temperature, catalyst concentration and type of catalyst. Each factor was used at two levels.

Factor	Levels
Temperature	160°C, 180°C
Catalyst concentration	20%, 40%
Type of catalyst	A, B

A factorial study was carried out in duplicate, and the average yield was determined.

Temperature	Concentration	Catalyst	Average yield
160°C	20%	A	60
160°C	20%	B	52
160°C	40%	A	54
160°C	40%	B	45
180°C	20%	A	72
180°C	20%	B	83
180°C	40%	A	68
180°C	40%	B	80

The estimated standard deviation of the experimental error was found to be 2.828.

Construct a table showing the effects representation of the three two-level factors in this process and their interactions. Calculate the main effects of the factors and interactions. Is there an interaction between any two factors in this chemical process?

Problem 8: An experiment is to be designed to study the corrosion rate of a reactor at a chemical acid plant. There are six factors of interest, each having two levels:

Factor	Levels
Raw-material feed rate	3000, 6000 pph
Gas temperature	100°C, 200°C
Scrubber water	5%, 20%
Reactor-bed acid	20%, 30%
Exit temperature	300°C, 360°C
Reactant distribution point	East, west

To minimize maintenance and downtime, it is desirable to operate the plant under process conditions that produce a low corrosion rate.

If all possible combinations of levels of the six process variables are to be tested, how many test runs would be required? Is it possible to design a small screening experiment to identify gross factor effects?

Problem 9: Consider the following two-level, full factorial design for three factors:

Run	A	B	C	y
1	-1	-1	-1	33
2	-1	-1	+1	57
3	-1	+1	-1	41
4	-1	+1	+1	59
5	+1	-1	-1	63
6	+1	-1	+1	51
7	+1	+1	-1	57
8	+1	+1	+1	53

Calculate the main effects of the factors. If only four randomly selected runs are carried out, what will be the calculated main effects of the three factors?

Problem 10: Calculate the main and interaction effects if a half-factorial design for the data given in Problem 9 is implemented.

Problem 11: Seven factors are believed to control the softness of cold-foamed car seats, and an experiment is to be conducted to determine how these factors influence softness. Only eight experiments are to be conducted due to time constraints.

Design an experiment to screen the effect of the seven factors (two levels each) on average softness of the seats.